

# Peer Review: From Chain-Ladder to Individual Claims Reserving

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**Authors:** Ronald Richman, Mario V. Wüthrich **arXiv:** 2602.15385v2 (Revised February 19, 2026)

## 1.1 Summary

This paper addresses the longstanding challenge of applying machine learning (ML) to individual (micro-level) claims reserving in non-life insurance — a domain where aggregate methods like the chain-ladder (CL) remain dominant due to their simplicity and robustness. The central research question is: can the CL method be restructured so that it naturally generalizes to individual claims ML models? The authors’ key contribution is a reformulation of CL factor estimation into recursive “projection-to-ultimate” (PtU) factors, which project the latest diagonal of a development triangle directly to ultimate claims, rather than chaining one-period-ahead steps forward. They prove (Proposition 2.2) that this backward extrapolation yields *identical* reserves to the classic CL predictor, creating a bridge to individual claims ML methods. As a proof of concept, feed-forward neural networks (FNNs) are applied to two real datasets — accident and liability insurance — demonstrating improved reserve accuracy over the standard CL benchmark.

## 1.2 Strengths

### 1.2.1 Elegant Conceptual Contribution

The core insight — that CL factors can be estimated recursively backward through the triangle, projecting observations directly to ultimates rather than chaining forward — is simple yet powerful. Proposition 2.2, which establishes exact equivalence between the two approaches at the aggregate level, is mathematically tight and rigorously proven in the appendix. This equivalence is the conceptual bridge that makes the extension to individual claims data natural and principled.

### 1.2.2 Resolution of a Key Barrier in Individual Claims Reserving

The paper correctly identifies the multi-period forecasting problem as the principal obstacle in prior ML-based individual claims reserving. By reformulating the problem as a one-shot ultimate prediction (option 4 in the taxonomy of Section 1), the method avoids the need to extrapolate stochastic dynamic covariates — a critical and often-overlooked bottleneck. This makes the approach genuinely practical with real-world claim feature data (incurred losses, claim status, case reserves, etc.).

### 1.2.3 Practical Guardrails via Chain-Ladder Regularization

A notable practical advantage emphasized throughout is that because the method is grounded in the CL framework, classical CL predictions can serve as guardrails to regularize ML forecasts. This is directly relevant to actuarial practice, where regulatory scrutiny and interpretability requirements make pure black-box ML models difficult to deploy.

### 1.2.4 Real Data Validation with Ground Truth

Both examples use datasets where the lower triangle is known, allowing the authors to benchmark forecasts against true outstanding loss liabilities (OLL). This is substantially stronger than backtesting on synthetic or simulated data, and the comparisons across accident insurance and liability insurance (Table 6 and Table 8) provide credible evidence of the method’s effectiveness.

### **1.2.5 Clear Roadmap for Future Development**

Section 5 provides a candid, detailed list of known limitations and next steps, including handling zero claims, lifting the Markov assumption with a Transformer architecture, cash-flow (sequence-to-sequence) forecasting, and IBNR integration. This transparency helps readers understand the current stage of the work and frames it appropriately as a foundational contribution.

## **1.3 Some Things to Look**

### **1.3.1 Loss Function and Zero Claims**

The use of mean squared error (MSE) as the loss function is acknowledged by the authors themselves (Section 5) as suboptimal, particularly for heavy-tailed liability claims and zero-payment claims. The exponential output activation precludes predictions of zero, which is a significant modeling gap for lines with high proportions of no-payment closures. This limits the practical utility of the current implementation for realistic portfolios.

### **1.3.2 IBNR Claims Excluded**

The method explicitly handles only RBNS claims, with IBNR reserves to be addressed by a downstream frequency-severity model (Section 4, limitation noted in Section 5). For many lines of business — especially long-tailed liability — IBNR can constitute the majority of total reserves. The paper’s scope is therefore materially narrower than a full reserving solution, and readers should be cautious about interpreting aggregate benchmark tables (e.g., Table 5) without this context. The paper could be clearer in its abstract and introduction about this restriction.

### **1.3.3 Sensitivity to Hyper-Parameters Not Fully Explored**

The authors specify hyper-parameters in Table 4 (batch size, learning rate, patience, etc.) and use an ensemble of 10 seeds, but no ablation studies or sensitivity analyses over these choices are provided. Given the small dataset sizes, the results could be sensitive to these decisions, and it is unclear how much the ensembling (nagging predictor technique, [16]) contributes to performance relative to the core methodological innovation.